

SYED MUHAMMAD HAMZAH RIZVI

Solutions Engineer | Applied AI

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PROFESSIONAL SUMMARY

Applied AI engineer with 3+ years working directly with customer technical contacts to architect integrations, run technical demos and scale solutions across up to 15 concurrent deployments spanning 10+ countries. Built and shipped a self-service AI support assistant now serving 300+ customer accounts weekly through a public embeddable widget, and authored 30+ integration guides that let customers implement independently. Partners closely with sales and account teams to strengthen the technical side of the sales process, and spots integration patterns worth productising back into the core platform. Strong Python background with deep exposure to REST APIs, webhooks, authentication flows and system integration patterns, paired with the communication range to move between engineers, operators and executives without losing either audience.

CORE SKILLS

- **Solutions Engineering & Pre-Sales:** Technical demos, solution architecture, customer discovery, sales enablement, self-service documentation, productisation of recurring integration patterns.
- **Applied AI:** LLMs, agentic workflows, RAG, semantic search, prompt design, grounding and hallucination reduction, model evaluation.
- **Integration & Engineering:** Python, C++, FastAPI, REST API design and debugging, webhooks, authentication flows, JSON, software architecture and integration-pattern design, Git.
- **Communication:** Bridging technical and non-technical stakeholders, tailoring messaging across engineering, sales and executive audiences, technical enablement content.

PROFESSIONAL EXPERIENCE

Technical Software Consultant — Innovative Technology, Manchester, UK

Mar 2023 – Present

- Work directly with customer technical contacts to architect integrations between AI-powered biometric and identity-verification products and their existing platforms, providing bespoke, hands-on support through go-live.
- Craft technical demonstrations and proof-of-concept walkthroughs that communicate product value clearly to both technical and business stakeholders during evaluation and rollout.
- Build reusable technical assets — integration guides, API documentation and self-service resources — that let customers implement and troubleshoot independently, reducing support load per account.
- Identify recurring integration patterns across customer deployments and feed them back to product and engineering as candidates for standardisation.
- Design and debug REST API integrations, webhooks and authentication flows connecting AI products into customer platforms and business systems.
- Diagnose complex production issues spanning APIs, devices, mobile apps, firmware and networks, translating technical root causes into clear explanations for non-technical stakeholders.
- Delivered rollouts across retail, gaming and compliance-focused environments spanning thousands of operational endpoints.

Automation Engineer — Tripack Films Ltd., Pakistan

Mar 2020 – Sep 2021

- Built and maintained industrial automation workflows, integrating control systems and sensors; diagnosed hardware and software issues in real-time operational environments.

Software Consultant — Evolligence, Pakistan

Dec 2019 – Mar 2020

- Built and supported backend application logic using structured APIs and data-driven workflows, collaborating with development teams on delivery.

SELECTED AI PROJECTS

GroundedOps — Production RAG Support Assistant — *Python, FastAPI, React, ChromaDB, Ollama, Docker, DeepSeek/OpenAI/Claude APIs*

- Deployed to production and now serving 300+ customer accounts weekly through a public embeddable widget and admin console; associated with a 40% reduction in first-line ticket volume and a 55% reduction in resolution time per internal Jira metrics since release (tracked alongside other concurrent initiatives, so not isolated to this tool alone).
- Built a retrieval-augmented QA assistant using hybrid retrieval (BM25 + dense embeddings), sigmoid-calibrated cross-encoder reranking and NLI-based grounding checks to reduce hallucination — an end-to-end example of architecting and shipping a working AI solution independently.

- Replaced fragile follow-up heuristics with an LLM-based Rewrite-Retrieve-Read query flow and per-session memory; built a 140+ test regression suite enabling fast, evidence-driven iteration.

Age Estimation & Demographic Analysis — *TensorFlow/Keras, EfficientNetB0, VGG16*

- Trained a hierarchical multi-task EfficientNetB0 model on 274k+ facial images (WIKI/IMDB, fine-tuned on UTKFace) with full data pipelines and evaluation tooling; achieved 87.5% gender accuracy, 88.4% coarse-age accuracy, 92.6% near-group (± 1) accuracy and 8.52-year MAE.

EDUCATION

MSc Robotics and Automation (Minor in AI) — University of Salford, UK

2021 – 2022

Graduated with Distinction. Dissertation: facial age and gender estimation using CNNs.

BEng Electrical Engineering — NUST, Pakistan

2015 – 2019

Focus: computer vision, machine learning and robotics.